

TATEYAMA, Japan

WMO: 476720

Lat: 34.9869N

Lon: 139.8653E

Elev: 7

StdP: 101.24

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	-0.5	-11.5	1.4	5.3	-9.6	1.7	5.5	11.3	10.5	10.2	10.3	1.8	160	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	31.7	25.9	30.8	25.7	30.0	25.4	26.8	29.9	26.4	29.3	26.0	28.8	4.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.3	28.5	25.5	20.8	28.2	25.1	20.2	28.0	84.2	30.0	82.3	29.5	80.4	28.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.3	7.3	-3.9	33.6	1.0	0.9	-4.6	34.3	-5.1	34.8	-5.7	35.4	-6.4	36.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-5.0	27.5	1.0	0.6	-5.6	27.9	-6.2	28.3	-6.8	28.6	-7.5	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	6.1	6.7	9.7	14.2	18.5	21.6	25.6	27.2	24.0	18.9	13.5	8.8
	DBStd	7.61	2.43	2.83	3.21	3.14	2.27	2.11	2.23	1.89	2.79	2.78	2.95	2.89
	HDD10.0	336	123	100	45	4	0	0	0	0	0	4	59	
	HDD18.3	1592	378	326	266	126	25	2	0	0	1	27	146	297
	CDD10.0	2637	4	8	37	132	264	349	485	532	419	274	110	21
	CDD18.3	851	0	0	0	3	32	101	227	274	170	43	2	0
	CDH23.3	6337	0	0	0	1	52	325	1824	2741	1237	156	1	0
	CDH26.7	1672	0	0	0	0	2	26	462	889	283	10	0	0
Wind	WSAvg	3.0	3.0	2.9	3.3	3.5	3.3	3.0	3.4	3.2	2.9	2.7	2.6	2.9
Precipitation	PrecAvg	1917	90	99	163	166	171	221	176	122	229	236	154	90
	PrecMax	2382	228	189	293	320	304	424	456	346	463	646	472	258
	PrecMin	1268	1	19	57	83	85	88	4	2	68	51	44	5
	PrecStd	303	55	47	59	61	56	88	122	91	101	142	94	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	18.4	20.2	23.0	26.2	29.1	32.5	33.2	31.5	28.0	23.1	20.3
		MCWB	13.8	14.9	14.9	17.7	19.4	23.5	26.1	26.4	25.2	22.4	18.4	17.4
	2%	DB	15.0	16.4	18.6	21.7	24.6	27.2	31.2	32.1	30.4	26.3	21.7	18.1
		MCWB	11.1	13.1	14.3	17.1	19.1	22.5	25.7	26.0	25.2	21.8	17.7	14.1
	5%	DB	13.3	14.5	17.4	20.7	23.7	26.1	30.3	31.3	29.4	25.1	20.7	16.5
		MCWB	8.8	10.8	13.6	16.3	18.7	21.8	25.4	25.9	24.8	21.2	16.7	12.0
	10%	DB	11.8	12.8	16.2	19.7	22.8	25.2	29.3	30.5	28.4	23.9	19.5	14.7
		MCWB	7.1	8.5	12.3	15.5	18.4	21.5	25.1	25.7	24.3	19.9	15.6	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	16.0	17.0	20.4	22.3	24.9	27.0	27.6	26.6	24.5	20.4	18.3
		MCDB	16.2	17.6	18.2	21.5	23.3	27.4	30.6	31.3	29.4	26.1	21.8	19.6
	2%	WB	11.7	13.8	15.6	18.6	21.1	23.9	26.4	27.1	26.1	23.1	18.8	14.9
		MCDB	14.4	15.8	17.7	20.4	22.9	25.9	29.6	30.3	28.7	25.3	20.9	17.3
	5%	WB	9.5	11.4	14.4	17.5	20.2	23.2	26.0	26.7	25.6	21.9	17.6	12.6
		MCDB	12.6	14.1	16.9	19.7	22.3	25.1	29.0	29.7	28.2	24.4	19.8	15.6
	10%	WB	7.7	9.2	12.8	16.4	19.5	22.4	25.5	26.3	25.1	20.6	16.2	10.9
		MCDB	11.1	12.1	15.6	18.9	21.8	24.3	28.3	29.2	27.7	23.2	18.7	13.9
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.1	8.8	8.4	7.4	6.0	5.8	6.3	6.8	7.7	8.7	9.4
		MCDBR	10.8	10.6	9.3	9.0	8.5	7.2	6.9	6.9	7.1	8.3	9.8	10.6
		MCWBR	8.1	8.5	7.0	5.5	4.5	3.5	2.8	2.6	3.3	5.1	6.6	7.8
	5% WB	MCDBR	10.0	9.7	8.5	6.8	6.6	5.3	6.2	6.1	6.3	7.1	8.1	9.7
		MCWBR	8.3	8.7	7.7	5.4	4.5	3.2	2.7	2.6	3.4	5.3	6.6	8.2
Clear-Sky Solar Irradiance	taub		0.388	0.438	0.500	0.543	0.557	0.572	0.568	0.550	0.499	0.482	0.432	0.389
	taud		2.233	2.090	1.941	1.882	1.910	1.948	2.004	2.046	2.160	2.121	2.178	2.237
	Ebn at Noon		791	787	770	759	754	741	741	745	765	735	734	761
	Edh at Noon		109	141	177	197	194	187	176	166	141	135	115	103
All-Sky Solar Radiation	RadAvg		2.62	3.11	3.97	4.70	5.12	4.50	5.17	5.25	4.03	3.02	2.44	2.33
	RadStd		0.21	0.35	0.28	0.49	0.54	0.45	0.84	0.44	0.35	0.32	0.22	0.20

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page